

# The Diameter of Acyclic and Preorder Preference Spaces under the Top-Difference Semimetric

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## Abstract

Let  $X$  be a finite set with  $|X| = n \geq 2$ , and let  $D$  be the top-difference semimetric on acyclic preference relations, defined by summing over all menus  $S \subseteq X$  the cardinality of the symmetric difference between the two sets of maximal elements. We prove that the diameter of the space  $A(X)$  of all acyclic orders and the diameter of the subspace  $P(X)$  of all preorders are equal and are given by

$$\text{diam}_D(A(X)) = \text{diam}_D(P(X)) = n2^{n-1} + 2 - 2^{\lfloor n/2 \rfloor} - 2^{\lceil n/2 \rceil}.$$

The lower bound is attained already by a pair of total preorders. The upper bound follows from a vertexwise decomposition of  $D$ , a sharp lower bound on the vertexwise deficit, and an elementary extremal fact about two topological orderings.

## 1 Introduction

Nishimura and Ok introduced a family of choice-behavioral semimetrics for preference relations, with the top-difference semimetric as the basic counting-measure case [1]. For a finite set  $X$  and two preference relations  $R, Q$ , this semimetric compares, menu by menu, the maximal elements selected by  $R$  and  $Q$ . The open problem considered here asks for the exact diameter of the space  $A(X)$  of acyclic orders and of the subspace  $P(X)$  of preorders under this semimetric [2].

The result is that the known lower bound is sharp. In fact, the extremal value is attained by two total preorders with complementary top indifference classes.

## 2 Definitions and notation

Let  $X$  be a finite set with  $|X| = n \geq 2$ . For a binary relation  $R$  on  $X$ , define its strict part by

$$y \succ_R x \iff yRx \text{ and not } xRy.$$

An *acyclic order* is a reflexive binary relation  $R$  whose strict part  $\succ_R$  has no directed cycle. Let  $A(X)$  denote the set of acyclic orders on  $X$ . Let  $P(X) \subseteq A(X)$  denote the set of preorders on  $X$ , that is, reflexive and transitive relations. The inclusion  $P(X) \subseteq A(X)$  is standard: the strict part of a preorder is transitive and asymmetric, hence acyclic.

For  $S \subseteq X$ , define the set of  $R$ -maximal elements of  $S$  by

$$M(S, R) := \{x \in S : \text{there is no } y \in S \text{ with } y \succ_R x\}.$$

The empty set and singletons cause no difficulty; they contribute zero to the distance below. The top-difference semimetric is

$$D(R, Q) := \sum_{S \subseteq X} |M(S, R) \triangle M(S, Q)|,$$

where  $\triangle$  denotes symmetric difference. For  $\mathcal{C} \subseteq A(X)$ , put

$$\text{diam}_D(\mathcal{C}) := \max\{D(R, Q) : R, Q \in \mathcal{C}\}.$$

For  $x \in X$ , write

$$N_R(x) := \{y \in X \setminus \{x\} : y \succ_R x\}$$

for the set of strict dominators of  $x$  under  $R$ .

### 3 A vertexwise decomposition

The first observation is that  $D(R, Q)$  decomposes into independent contributions from the individual alternatives.

**Lemma 1** (Vertexwise formula). *Let  $R, Q \in A(X)$ , let  $x \in X$ , and set*

$$A_x := N_R(x), \quad B_x := N_Q(x),$$

with

$$a_x := |A_x|, \quad b_x := |B_x|, \quad u_x := |A_x \cup B_x|.$$

Let  $m = n - 1$ . Define  $d_x(R, Q)$  to be the number of subsets  $S \subseteq X$  such that  $x$  belongs to exactly one of  $M(S, R)$  and  $M(S, Q)$ . Then

$$D(R, Q) = \sum_{x \in X} d_x(R, Q)$$

and

$$d_x(R, Q) = 2^{m-a_x} + 2^{m-b_x} - 2 \cdot 2^{m-u_x}.$$

Equivalently, if

$$\delta_x(R, Q) := 2^m - d_x(R, Q),$$

then

$$\delta_x(R, Q) = 2^m - 2^{m-a_x} - 2^{m-b_x} + 2 \cdot 2^{m-u_x}.$$

*Proof.* For any  $S \subseteq X$ , the contribution of  $x$  to  $|M(S, R) \triangle M(S, Q)|$  is the absolute difference between the two indicators

$$\mathbf{1}_{\{x \in M(S, R)\}}, \quad \mathbf{1}_{\{x \in M(S, Q)\}}.$$

Summing first over  $x$  and then over  $S$  gives

$$D(R, Q) = \sum_{x \in X} d_x(R, Q).$$

Fix  $x$ . Only subsets  $S$  containing  $x$  can contribute to  $d_x(R, Q)$ . Write  $T = S \setminus \{x\}$ , so  $T \subseteq X \setminus \{x\}$  and  $|X \setminus \{x\}| = m$ . By the definition of maximality,

$$x \in M(S, R) \iff T \cap A_x = \emptyset,$$

and similarly

$$x \in M(S, Q) \iff T \cap B_x = \emptyset.$$

Thus  $d_x(R, Q)$  is the number of  $T \subseteq X \setminus \{x\}$  for which exactly one of the two events

$$T \cap A_x = \emptyset, \quad T \cap B_x = \emptyset$$

holds. There are  $2^{m-a_x}$  subsets avoiding  $A_x$ ,  $2^{m-b_x}$  subsets avoiding  $B_x$ , and  $2^{m-u_x}$  subsets avoiding  $A_x \cup B_x$ . Inclusion-exclusion for the symmetric difference of the two events gives

$$d_x(R, Q) = 2^{m-a_x} + 2^{m-b_x} - 2 \cdot 2^{m-u_x},$$

as claimed. The displayed formula for  $\delta_x$  follows by subtracting from  $2^m$ .  $\square$

The next lemma is the key local inequality. It says that, no matter how the two strict dominator sets overlap, the deficit of the  $x$ -contribution from its trivial upper bound  $2^{n-1}$  is at least an exponential term determined by the larger of the two dominator sets.

**Lemma 2** (Local deficit bound). *With the notation of Lemma 1,*

$$\delta_x(R, Q) \geq 2^{m-\max\{a_x, b_x\}}.$$

*Proof.* Suppress  $x$  from the notation and write  $a = |A|$ ,  $b = |B|$ , and  $u = |A \cup B|$ . By symmetry, assume  $a \leq b$ . We must prove

$$2^m - 2^{m-a} - 2^{m-b} + 2 \cdot 2^{m-u} \geq 2^{m-b}.$$

After division by  $2^m$ , this is equivalent to

$$1 - 2^{-a} - 2^{1-b} + 2^{1-u} \geq 0.$$

Since  $u \leq a + b$ , we have  $2^{1-u} \geq 2^{1-a-b}$ , so it is enough to show

$$1 - 2^{-a} - 2^{1-b} + 2^{1-a-b} \geq 0.$$

But

$$1 - 2^{-a} - 2^{1-b} + 2^{1-a-b} = (1 - 2^{-a})(1 - 2^{1-b}).$$

This product is nonnegative. Indeed, if  $a = 0$ , the first factor is zero; if  $a \geq 1$ , then  $b \geq a \geq 1$ , so both factors are nonnegative. Therefore  $\delta_x(R, Q) \geq 2^{m-b} = 2^{m-\max\{a, b\}}$ , proving the lemma.  $\square$

## 4 An extremal fact about two topological orderings

We next record the elementary combinatorial estimate that controls the sum of the local lower bounds.

**Lemma 3** (Two-permutation estimate). *Let  $\pi, \sigma : X \rightarrow \{1, \dots, n\}$  be bijections, and define*

$$r_x := \max\{\pi(x), \sigma(x)\} \quad (x \in X).$$

*Then*

$$\sum_{x \in X} 2^{n-r_x} \geq 2^{\lfloor n/2 \rfloor} + 2^{\lceil n/2 \rceil} - 2.$$

*The bound is sharp.*

*Proof.* For  $k = 0, 1, \dots, n$ , put

$$C_k := |\{x \in X : r_x \leq k\}|.$$

Thus  $C_0 = 0$  and  $C_n = n$ . Also let

$$P_k := \{x : \pi(x) \leq k\}, \quad Q_k := \{x : \sigma(x) \leq k\}.$$

Then

$$\{x : r_x \leq k\} = P_k \cap Q_k.$$

Since  $|P_k| = |Q_k| = k$ , the union bound gives

$$C_k = |P_k \cap Q_k| = |P_k| + |Q_k| - |P_k \cup Q_k| \geq 2k - n.$$

Of course  $C_k \geq 0$ , hence

$$C_k \geq L_k := \max\{0, 2k - n\} \quad (0 \leq k \leq n).$$

Let  $f(k) = 2^{n-k}$ . Since  $C_k - C_{k-1}$  is the number of  $x$  with  $r_x = k$ , summation by parts yields

$$\sum_{x \in X} 2^{n-r_x} = \sum_{k=1}^n f(k)(C_k - C_{k-1}) = C_n f(n) + \sum_{k=1}^{n-1} C_k (f(k) - f(k+1)).$$

The coefficients  $f(k) - f(k+1)$  are positive, so the lower bounds  $C_k \geq L_k$  imply that the sum is minimized when the cumulative counts are  $L_k$ . Equivalently, the extremal multiset of possible  $r$ -values has cumulative distribution  $L_k$ .

If  $n = 2s$ , then  $L_k = 0$  for  $k \leq s$  and  $L_k = 2k - 2s$  for  $s < k \leq 2s$ . Hence the extremal distribution has two elements with each value  $k = s+1, s+2, \dots, 2s$ , and therefore

$$\sum_{x \in X} 2^{n-r_x} \geq 2 \sum_{k=s+1}^{2s} 2^{2s-k} = 2 \sum_{j=0}^{s-1} 2^j = 2^{s+1} - 2.$$

Since  $\lfloor n/2 \rfloor = \lceil n/2 \rceil = s$ , this equals  $2^{\lfloor n/2 \rfloor} + 2^{\lceil n/2 \rceil} - 2$ .

If  $n = 2s + 1$ , then  $L_k = 0$  for  $k \leq s$ ,  $L_{s+1} = 1$ , and thereafter  $L_k - L_{k-1} = 2$  for  $k = s + 2, \dots, 2s + 1$ . Hence

$$\sum_{x \in X} 2^{n-r_x} \geq 2^{2s+1-(s+1)} + 2 \sum_{k=s+2}^{2s+1} 2^{2s+1-k} = 2^s + 2 \sum_{j=0}^{s-1} 2^j = 3 \cdot 2^s - 2.$$

Since  $\lfloor n/2 \rfloor = s$  and  $\lceil n/2 \rceil = s + 1$ , this is again  $2^{\lfloor n/2 \rfloor} + 2^{\lceil n/2 \rceil} - 2$ .

Sharpness is witnessed by opposite orders: take  $X = \{1, \dots, n\}$ ,  $\pi(i) = i$ , and  $\sigma(i) = n + 1 - i$ . Then

$$|\{i : \max(i, n + 1 - i) \leq k\}| = \max\{0, 2k - n\} = L_k$$

for every  $k$ . □

## 5 The upper bound for acyclic orders

We now prove that no pair of acyclic orders can exceed the claimed value.

**Theorem 1** (Upper bound). *For every finite set  $X$  with  $|X| = n \geq 2$ ,*

$$\text{diam}_D(A(X)) \leq n2^{n-1} + 2 - 2^{\lfloor n/2 \rfloor} - 2^{\lceil n/2 \rceil}.$$

*Proof.* Let  $R, Q \in A(X)$  be arbitrary. Since the strict parts  $\succ_R$  and  $\succ_Q$  are acyclic directed graphs on the finite set  $X$ , choose topological orderings

$$\pi_R, \pi_Q : X \rightarrow \{1, \dots, n\}$$

such that

$$y \succ_R x \implies \pi_R(y) < \pi_R(x), \quad y \succ_Q x \implies \pi_Q(y) < \pi_Q(x).$$

For  $x \in X$ , put

$$a_x := |N_R(x)|, \quad b_x := |N_Q(x)|, \quad r_x := \max\{\pi_R(x), \pi_Q(x)\}.$$

Because every strict dominator of  $x$  under  $R$  must occur before  $x$  in  $\pi_R$ ,

$$a_x \leq \pi_R(x) - 1.$$

Similarly,

$$b_x \leq \pi_Q(x) - 1.$$

Therefore

$$\max\{a_x, b_x\} \leq r_x - 1.$$

By Lemma 2, with  $m = n - 1$ ,

$$\delta_x(R, Q) \geq 2^{m - \max\{a_x, b_x\}} \geq 2^{m - (r_x - 1)} = 2^{n - r_x}.$$

Summing over  $x$  and applying Lemma 3 to the two bijections  $\pi_R$  and  $\pi_Q$ , we get

$$\sum_{x \in X} \delta_x(R, Q) \geq \sum_{x \in X} 2^{n - r_x} \geq 2^{\lfloor n/2 \rfloor} + 2^{\lceil n/2 \rceil} - 2.$$

By Lemma 1,

$$D(R, Q) = \sum_{x \in X} d_x(R, Q) = \sum_{x \in X} (2^{n-1} - \delta_x(R, Q)) = n2^{n-1} - \sum_{x \in X} \delta_x(R, Q).$$

Consequently,

$$D(R, Q) \leq n2^{n-1} - (2^{\lfloor n/2 \rfloor} + 2^{\lceil n/2 \rceil} - 2) = n2^{n-1} + 2 - 2^{\lfloor n/2 \rfloor} - 2^{\lceil n/2 \rceil}.$$

Since  $R, Q \in A(X)$  were arbitrary, the asserted upper bound follows.  $\square$

## 6 An extremal pair of preorders

It remains to show that the upper bound is attained inside  $P(X)$ . The construction is explicit.

**Theorem 2** (Attainment in preorders). *For every finite set  $X$  with  $|X| = n \geq 2$ ,*

$$\text{diam}_D(P(X)) \geq n2^{n-1} + 2 - 2^{\lfloor n/2 \rfloor} - 2^{\lceil n/2 \rceil}.$$

*Proof.* Put

$$p := \lfloor n/2 \rfloor, \quad q := \lceil n/2 \rceil,$$

and partition  $X$  as

$$X = U \sqcup V, \quad U = \{u_1, \dots, u_q\}, \quad V = \{v_1, \dots, v_p\}.$$

Define a preorder  $R$  by the ordered indifference classes

$$U \succ_R \{v_1\} \succ_R \{v_2\} \succ_R \dots \succ_R \{v_p\},$$

meaning that all elements of  $U$  are tied with each other, every element of  $U$  is strictly above every element of  $V$ , and

$$v_1 \succ_R v_2 \succ_R \dots \succ_R v_p.$$

Define a preorder  $Q$  by

$$V \succ_Q \{u_1\} \succ_Q \{u_2\} \succ_Q \dots \succ_Q \{u_q\}.$$

These are total preorders, hence belong to  $P(X)$ .

We compute the deficits  $\delta_x(R, Q)$  from Lemma 1. For  $u_i$ , the set of strict dominators under  $R$  is empty, while the strict dominators under  $Q$  are

$$V \cup \{u_1, \dots, u_{i-1}\}.$$

Thus

$$|N_R(u_i)| = 0, \quad |N_Q(u_i)| = p + i - 1, \quad |N_R(u_i) \cup N_Q(u_i)| = p + i - 1.$$

With  $m = n - 1$ , Lemma 1 gives

$$\delta_{u_i}(R, Q) = 2^m - 2^m - 2^{m-(p+i-1)} + 2 \cdot 2^{m-(p+i-1)} = 2^{m-(p+i-1)} = 2^{q-i}.$$

Summing over  $i = 1, \dots, q$  gives

$$\sum_{i=1}^q \delta_{u_i}(R, Q) = \sum_{i=1}^q 2^{q-i} = 2^q - 1.$$

Similarly, for  $v_j$ , the set of strict dominators under  $Q$  is empty, while the strict dominators under  $R$  are

$$U \cup \{v_1, \dots, v_{j-1}\}.$$

Hence

$$\delta_{v_j}(R, Q) = 2^{p-j},$$

and therefore

$$\sum_{j=1}^p \delta_{v_j}(R, Q) = 2^p - 1.$$

The total deficit is

$$\sum_{x \in X} \delta_x(R, Q) = (2^q - 1) + (2^p - 1) = 2^q + 2^p - 2.$$

Using again

$$D(R, Q) = n2^{n-1} - \sum_{x \in X} \delta_x(R, Q),$$

we obtain

$$D(R, Q) = n2^{n-1} + 2 - 2^p - 2^q = n2^{n-1} + 2 - 2^{\lfloor n/2 \rfloor} - 2^{\lceil n/2 \rceil}.$$

Thus the claimed value is attained by two preorders.  $\square$

## 7 Conclusion

Combining the upper bound for all acyclic orders with the explicit preorder construction gives the exact diameters.

**Corollary 1.** *For every finite set  $X$  with  $|X| = n \geq 2$ ,*

$$\boxed{\text{diam}_D(A(X)) = \text{diam}_D(P(X)) = n2^{n-1} + 2 - 2^{\lfloor n/2 \rfloor} - 2^{\lceil n/2 \rceil}}.$$

*Moreover, the value is attained by a pair of total preorders.*

*Proof.* By Theorem 1, every pair of acyclic orders is at distance at most the displayed quantity. Since  $P(X) \subseteq A(X)$ , the same upper bound holds a fortiori on  $P(X)$ . By Theorem 2, the displayed quantity is attained by a pair of preorders. Hence both diameters are equal to that value.  $\square$

## References

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